OZ Lab

est. August 2024

#4 / Multiple Features and Feature Scaling

Gradient Descent for Linear Regression

@oz.lab__

Predict, Measure, and Update

Linear regression model

$$f_{w,b}(x) = wx + b$$

Cost function

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}(f_{w,b}(x^{(i)}) - y^{(i)})^2$$

Gradient descent algorithm

repeat until convergence {

$$w = w - \alpha\frac{\partial}{\partial w}J(w,b) \rightarrow \frac{1}{m}\sum_{i=1}^{m}(f_{w,b}(x^{(i)}) - y^{(i)})x^{(i)}$$

$$b = b - \alpha\frac{\partial}{\partial b}J(w,b) \rightarrow \frac{1}{m}\sum_{i=1}^{m}(f_{w,b}(x^{(i)}) - y^{(i)})$$

}

next slide is optional!

Where the Derivatives Come From

$$\frac{\partial}{\partial w} J(w,b) = \frac{d}{dw}\frac{1}{2m}\sum_{i=1}^{m}\left(f_{w,b}(x^{(i)}) - y^{(i)}\right)^2 = \frac{d}{dw}\frac{1}{2m}\sum_{i=1}^{m}\left(wx^{(i)} + b - y^{(i)}\right)^2$$

$$= \frac{1}{2m}\sum_{i=1}^{m}\left(wx^{(i)} + b - y^{(i)}\right)2x^{(i)} = \boxed{\frac{1}{m}\sum_{i=1}^{m}\left(f_{w,b}(x^{(i)}) - y^{(i)}\right)x^{(i)}}$$

$$\frac{\partial}{\partial b} J(w,b) = \frac{d}{db}\frac{1}{2m}\sum_{i=1}^{m}\left(f_{w,b}(x^{(i)}) - y^{(i)}\right)^2 = \frac{d}{db}\frac{1}{2m}\sum_{i=1}^{m}\left(wx^{(i)} + b - y^{(i)}\right)^2$$

$$= \frac{1}{2m}\sum_{i=1}^{m}\left(wx^{(i)} + b - y^{(i)}\right)2 = \boxed{\frac{1}{m}\sum_{i=1}^{m}\left(f_{w,b}(x^{(i)}) - y^{(i)}\right)}$$

no $x^{(i)}$

The Linear Regression Update Rule

$\frac{d}{dw} J(w,b)$

repeat until convergence {

$$w = w - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}\left(x^{(i)}\right) - y^{(i)} \right) x^{(i)}$$

$$b = b - \alpha \frac{1}{m} \sum_{i=1}^{m} \left(f_{w,b}\left(x^{(i)}\right) - y^{(i)} \right)$$

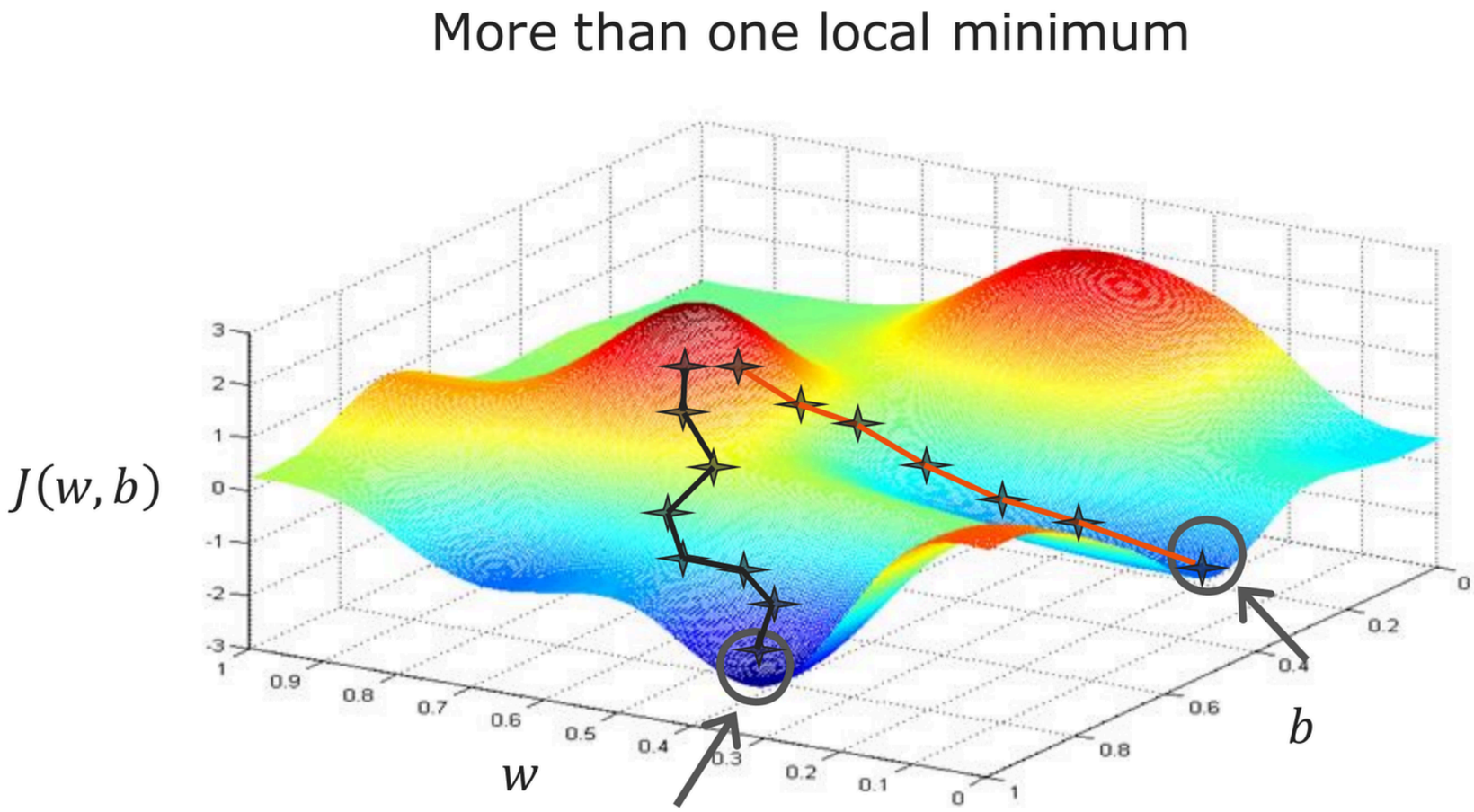
}

Update w and b simultaneously

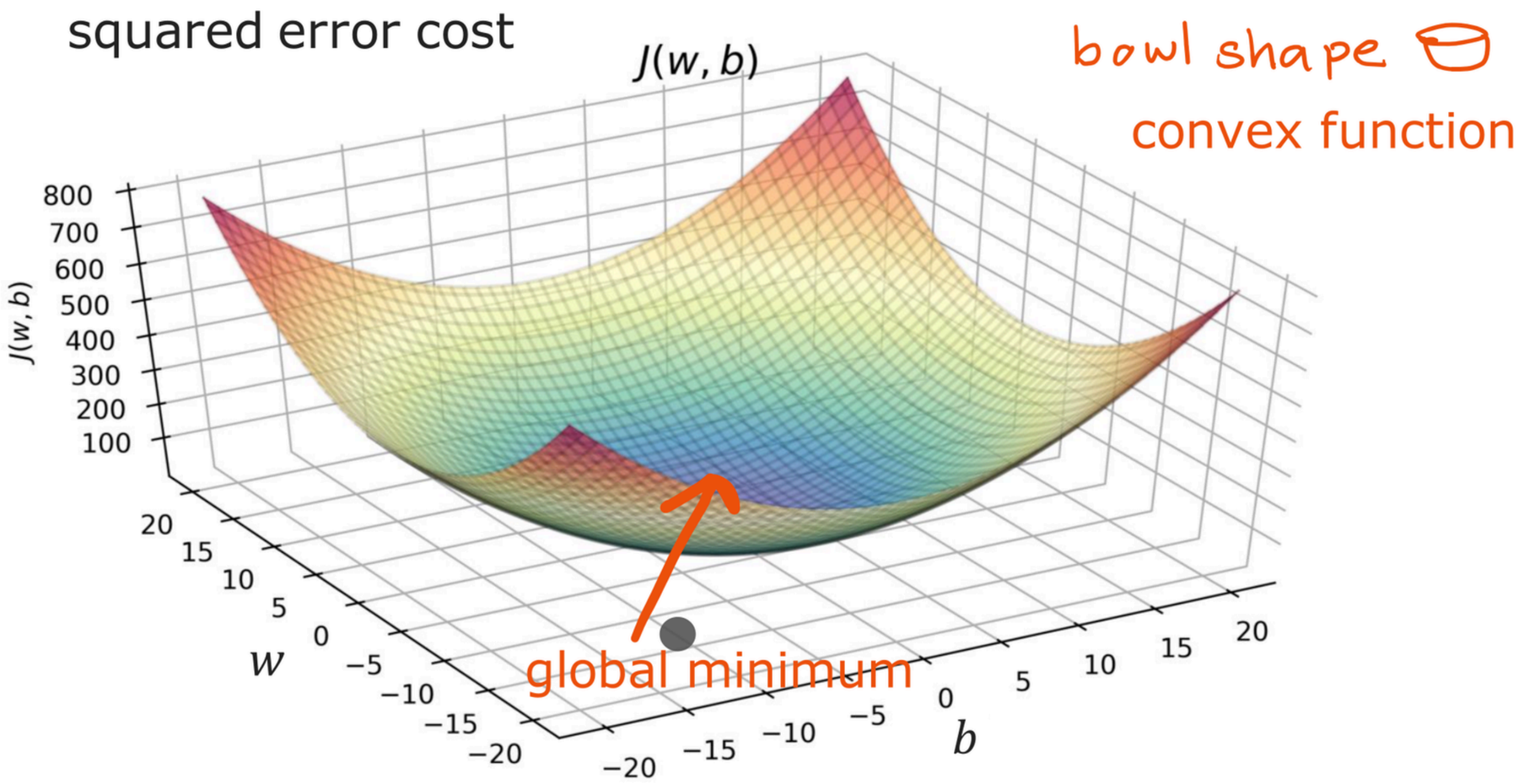
$\frac{d}{db} J(w,b)$

$f_{w,b}(x^{(i)}) = wx^{(i)} + b$

More than One Local Minimum



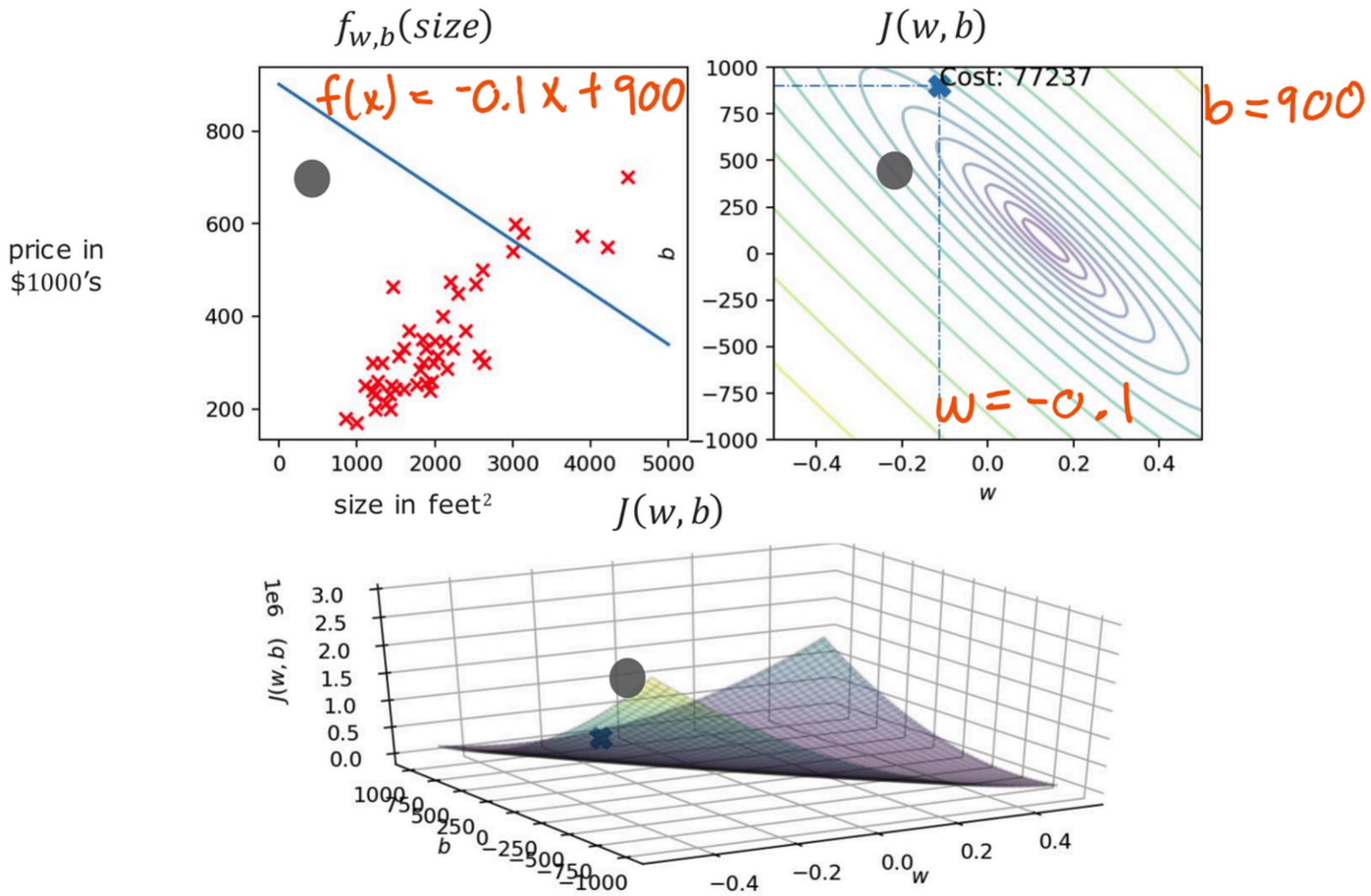
A Convex Cost and Its Global Minimum



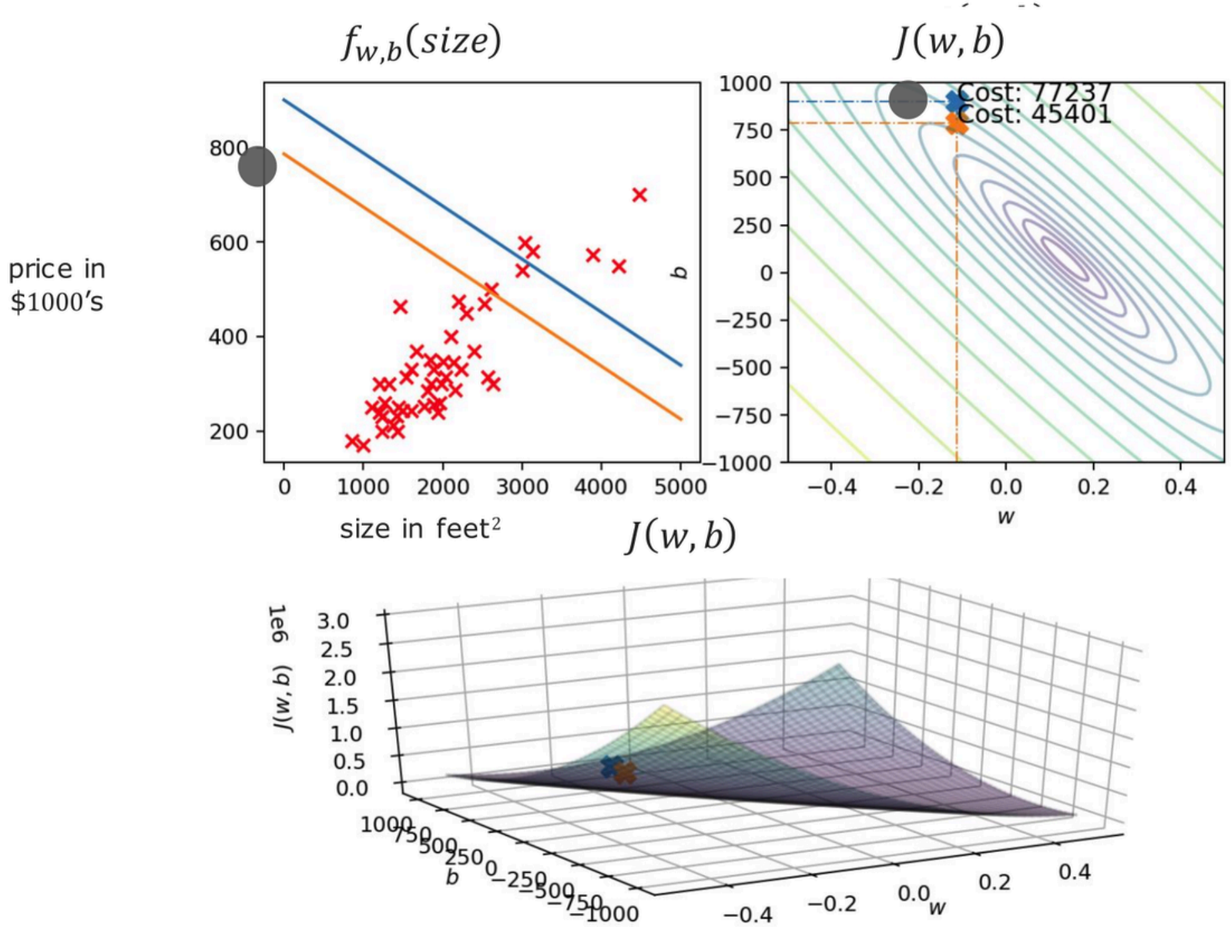
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Running Gradient Descent

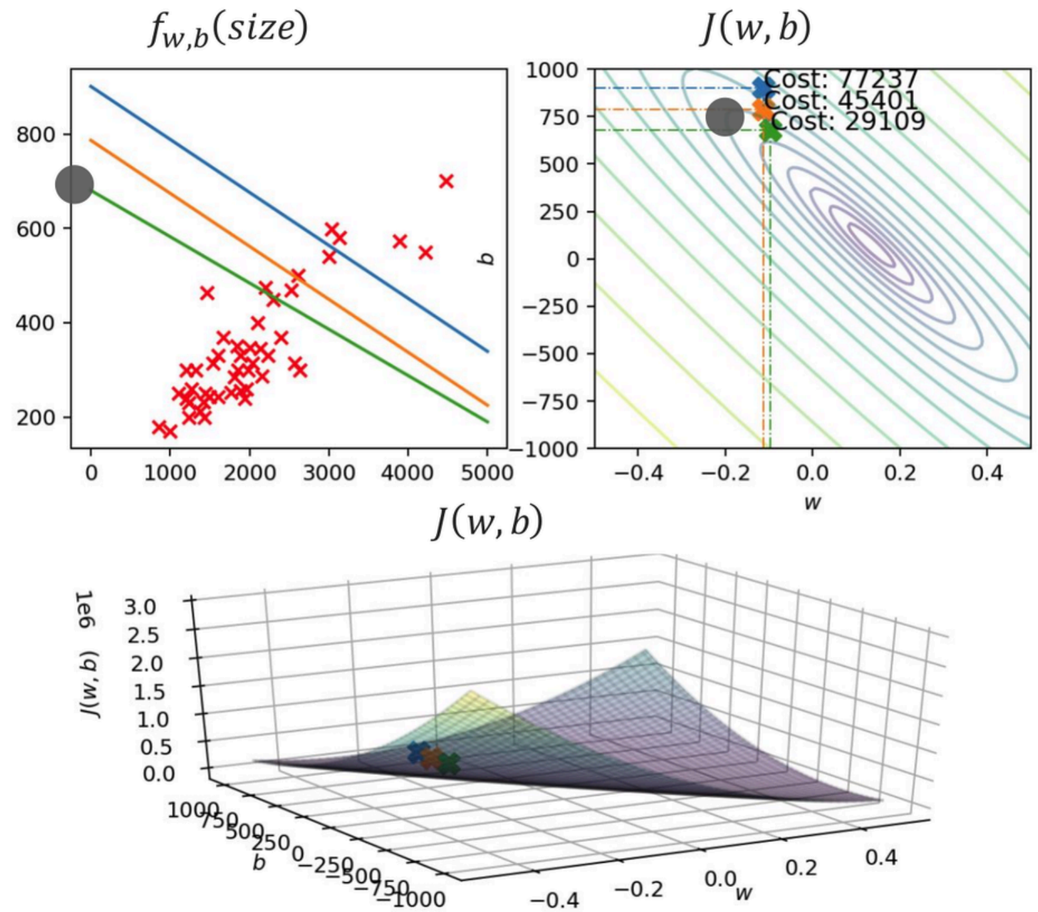
Starting with a Poor Fit



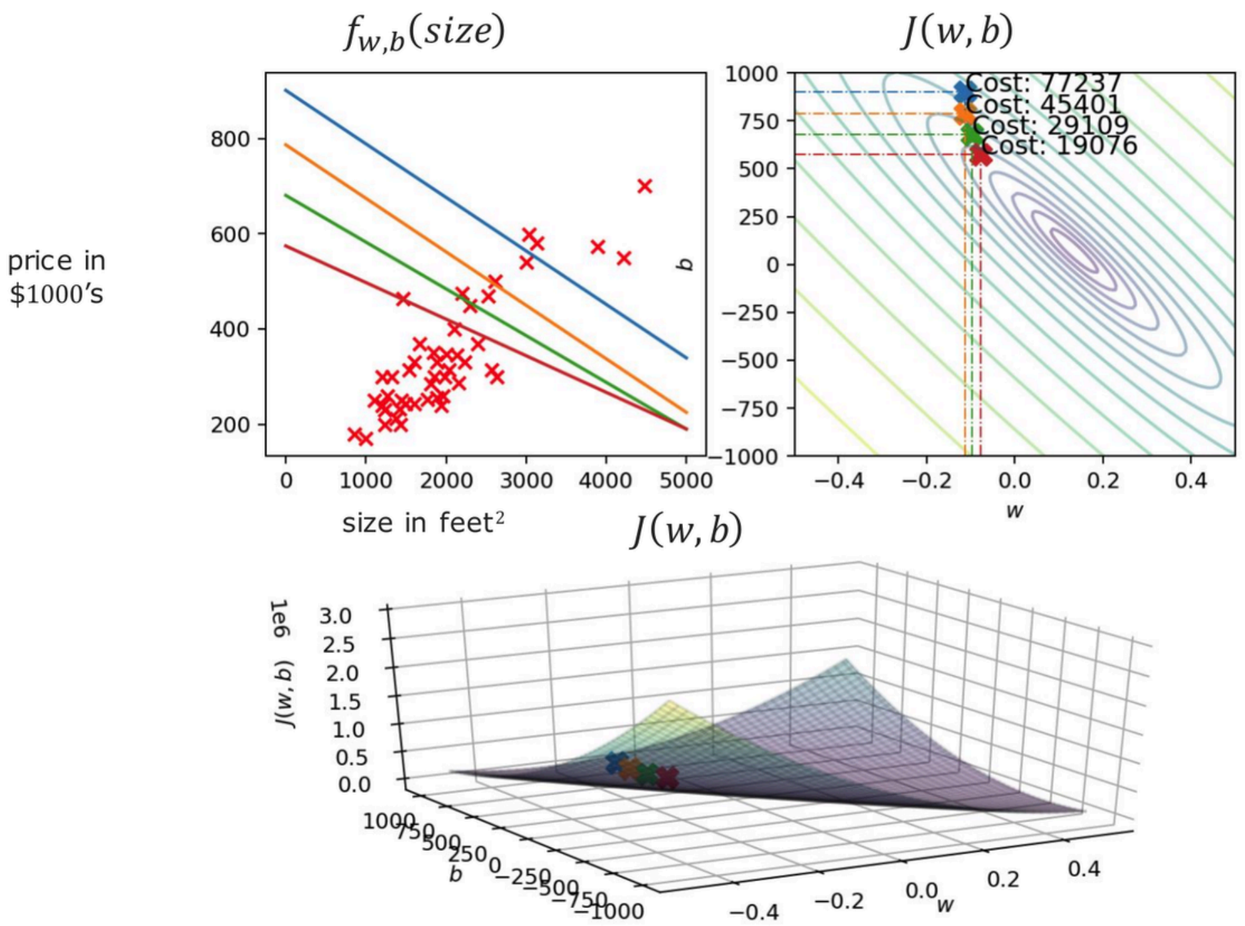
The First Gradient Descent Update



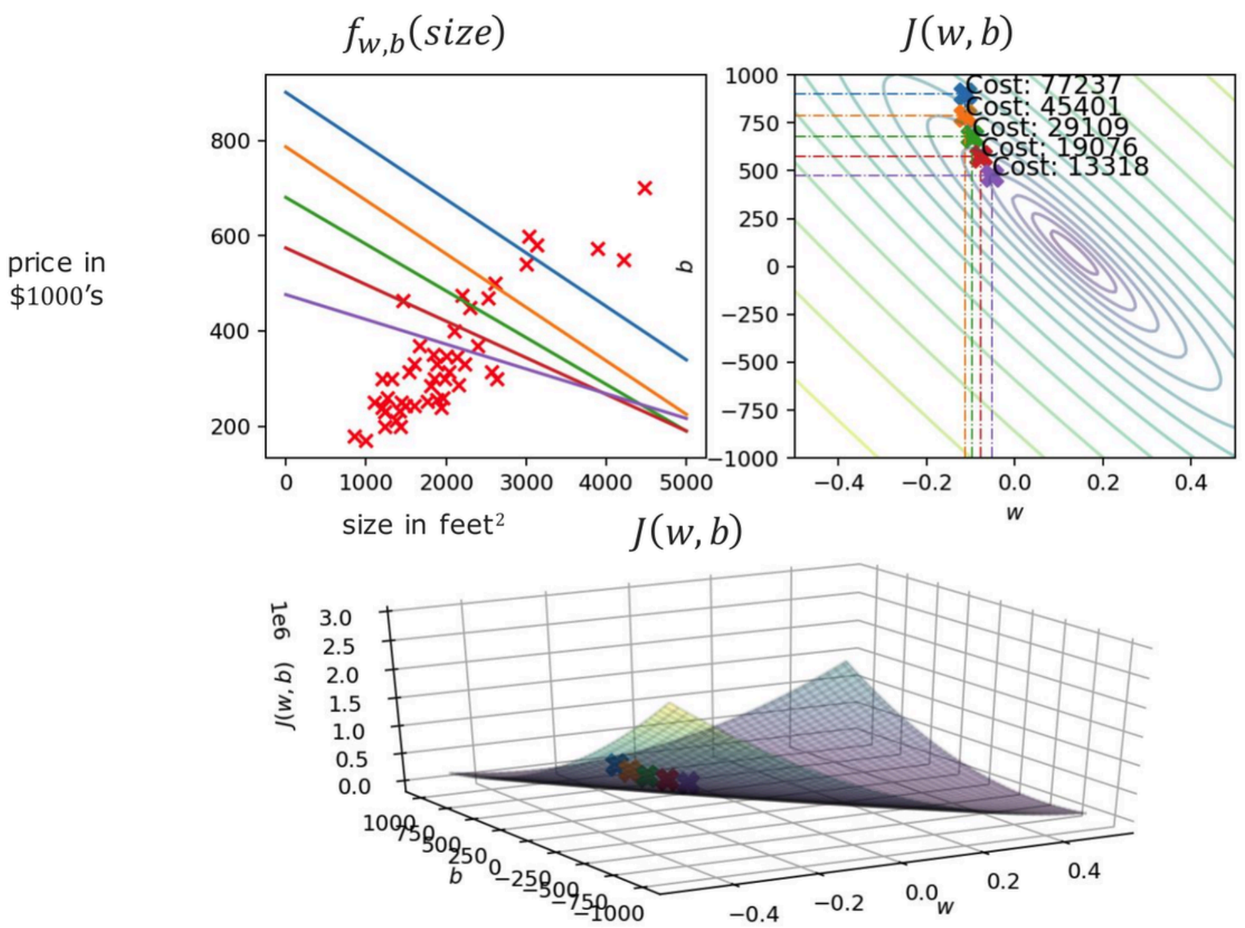
Lowering Cost Step by Step



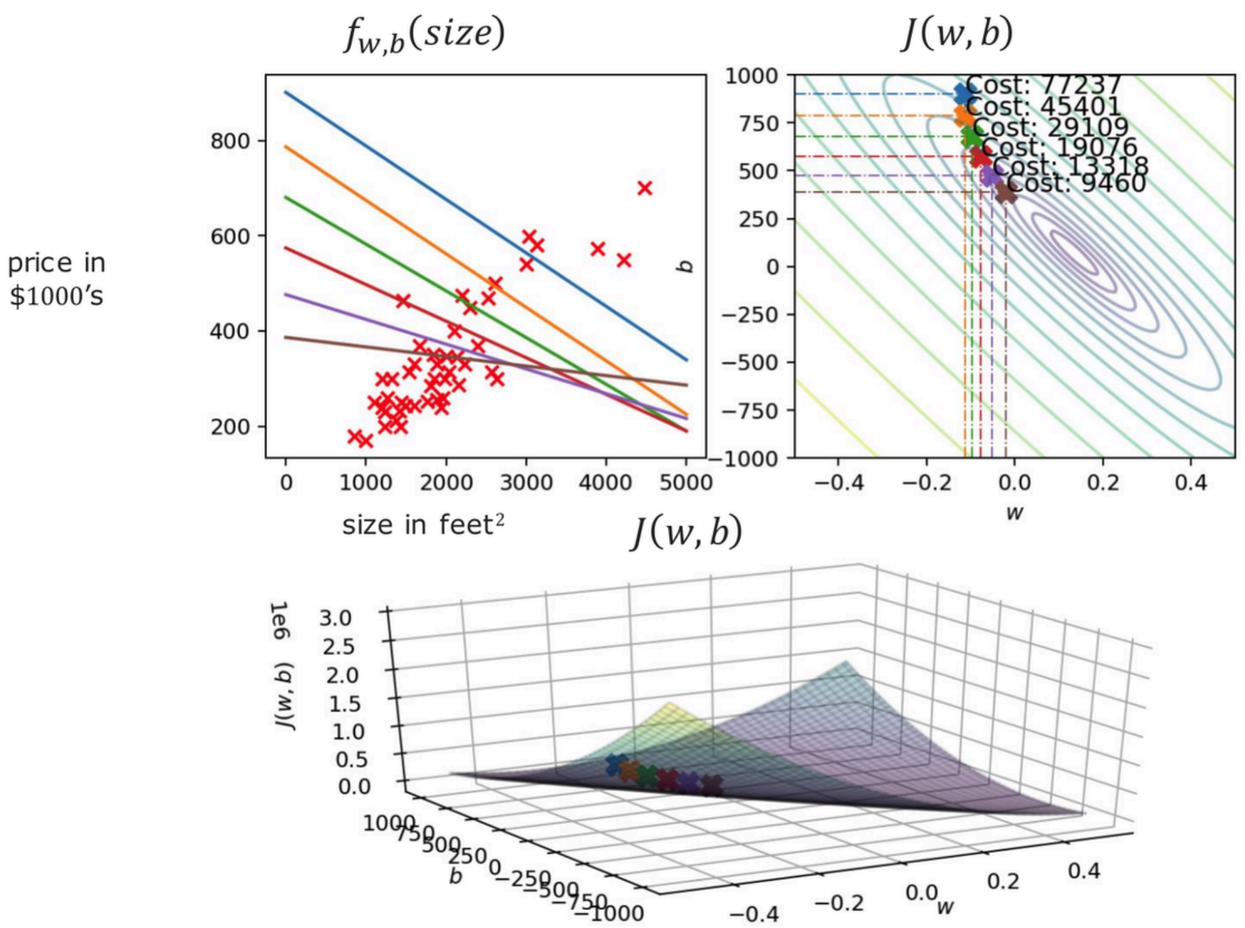
Changing the Slope and Intercept



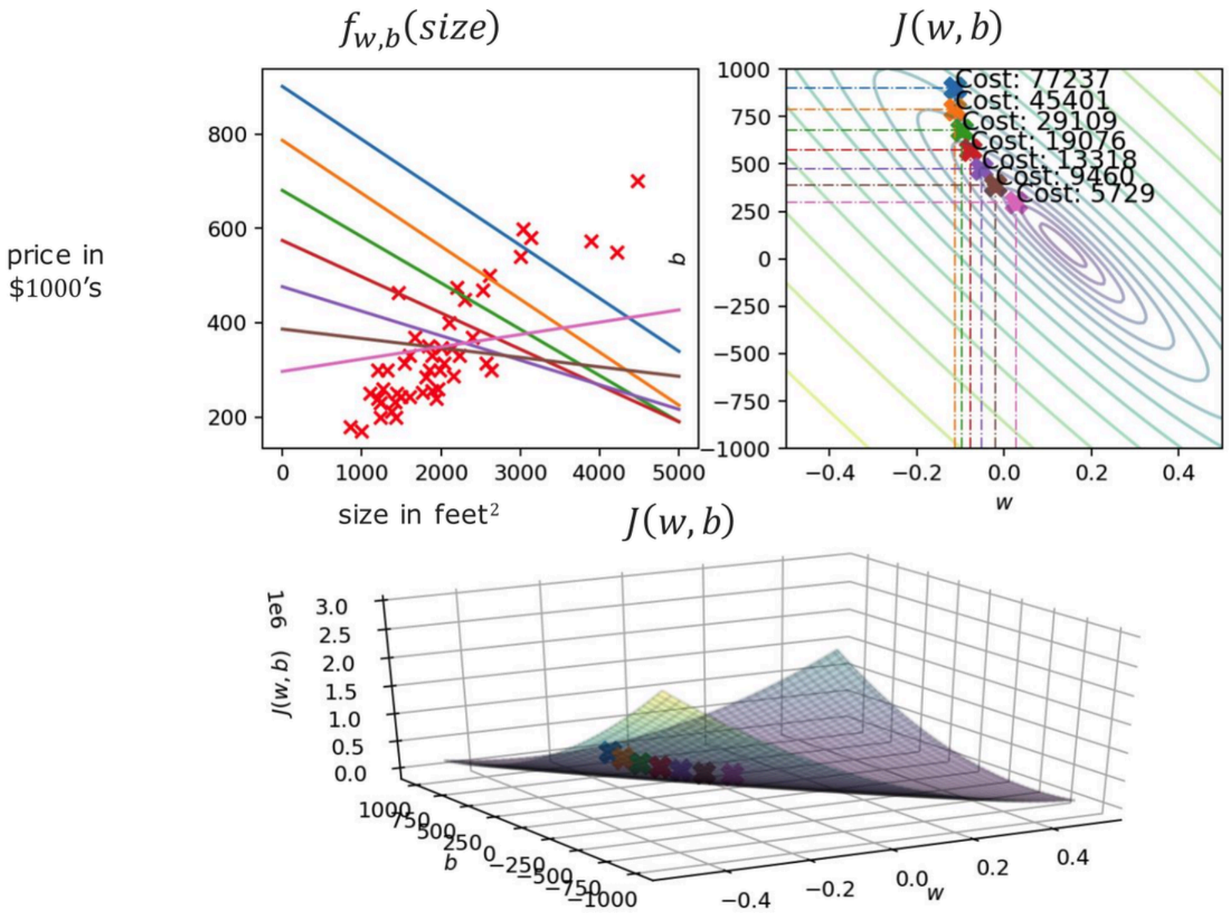
The Line Becomes Flatter



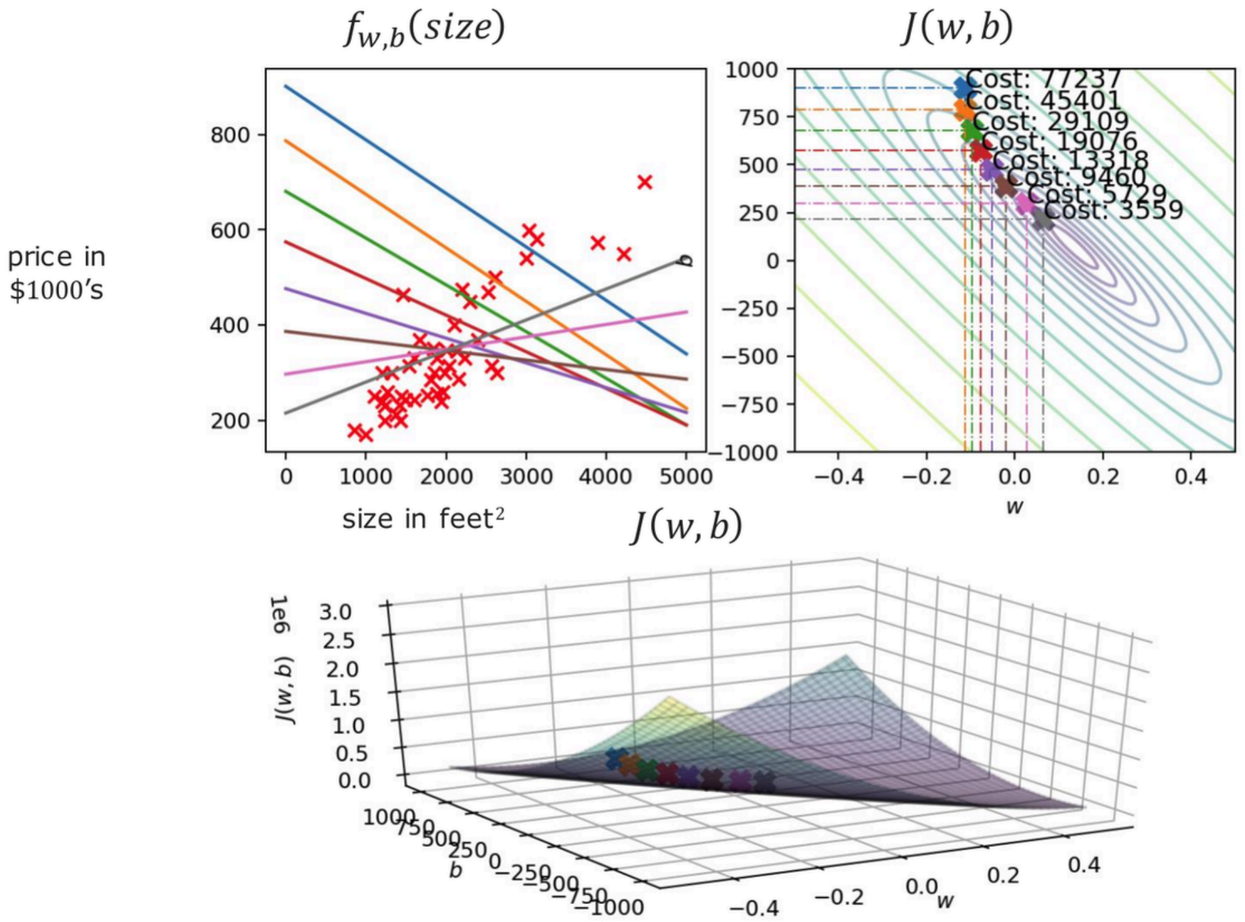
Approaching a Horizontal Line



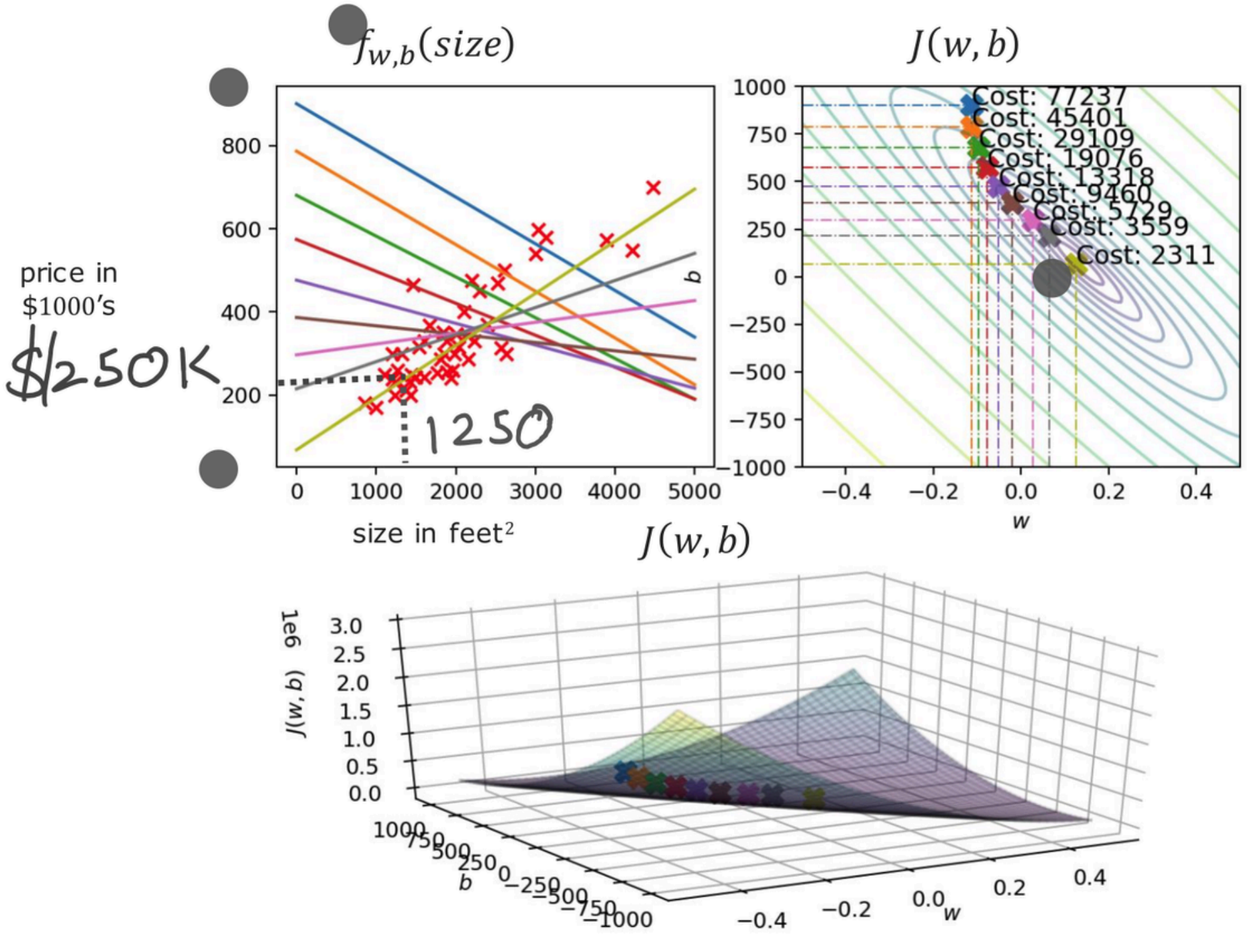
Finding a Positive Slope



Approaching the Best Fit



Predicting with the Learned Line



Learning from Every Training Example



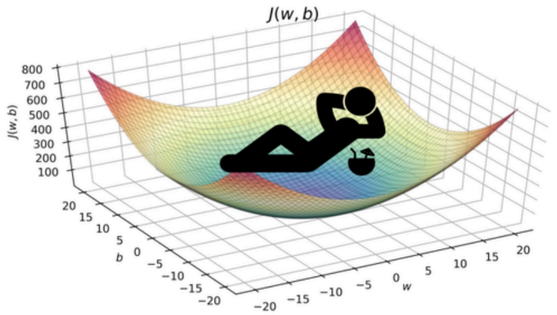
other gradient descent: subsets

"Batch": Each step of gradient descent uses all the training examples.

	x size in feet^2	y price in \$1000's
(1)	2104	400
(2)	1416	232
(3)	1534	315
(4)	852	178
...	...	...
(47)	3210	870

$m = 47$

$$\sum_{i=1}^{m} (f_{w,b}(x^{(i)}) - y^{(i)})^2$$



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Linear Regression with Many Features

Reviewing the Single Feature Model

one feature →

Size in feet^2 (x)	Price ($) in 1000's ($y$) ←
2104	400
1416	232
1534	315
852	178
...	...

$$f_{w,b}(x) = wx + b$$

Reading Rows, Features, and Examples

	Size in feet²	Number of bedrooms	Number of floors	Age of home in years	Price ($) in $1000's
	X_1	X_2	X_3	X_4	
	2104	5	1	45	460
$i=2$	1416	3	2	40	232
	1534	3	2	30	315
	852	2	1	36	178
	...	...	...	...	...

$j = 1 \ldots 4$

$n = 4$

- x_j = j^{th} feature
- n = number of features
- $\vec{x}^{(i)}$ = features of ith training example
- $x_j^{(i)}$ = value of feature j in i^{th} training example

- $\vec{x}^{(2)} = [1416 \;\; 3 \;\; 2 \;\; 40]$

$x_3^{(2)} = 2$

Giving Each Feature a Weight

Previously: $f_{w,b}(x) = wx + b$

$$f_{w,b}(X) = w_1 X_1 + w_2 X_2 + w_3 X_3 + w_4 X_4 + b$$

example

$$f_{w,b}(X) = 0.1 X_1 + 4 X_2 + 10 X_3 + -2 X_4 + 80$$

X_1: size
X_2: # bedrooms
X_3: # floors
X_4: years
80: base price

$$f_{w,b}(x) = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b$$

Writing the Model with Vectors

$$f_{\vec{w},b}(\vec{x}) = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b$$

$\vec{w} = [w_1 \; w_2 \; w_3 \ldots w_n]$ parameters of the model

b is a number

vector $\vec{x} = [x_1 \; x_2 \; x_3 \ldots x_n]$

$$f_{\vec{w},b}(\vec{x}) = \vec{w} \cdot \vec{x} + b = w_1 x_1 + w_2 x_2 + w_3 x_3 + \cdots + w_n x_n + b$$

dot product

multiple linear regression

(not multivariate regression)

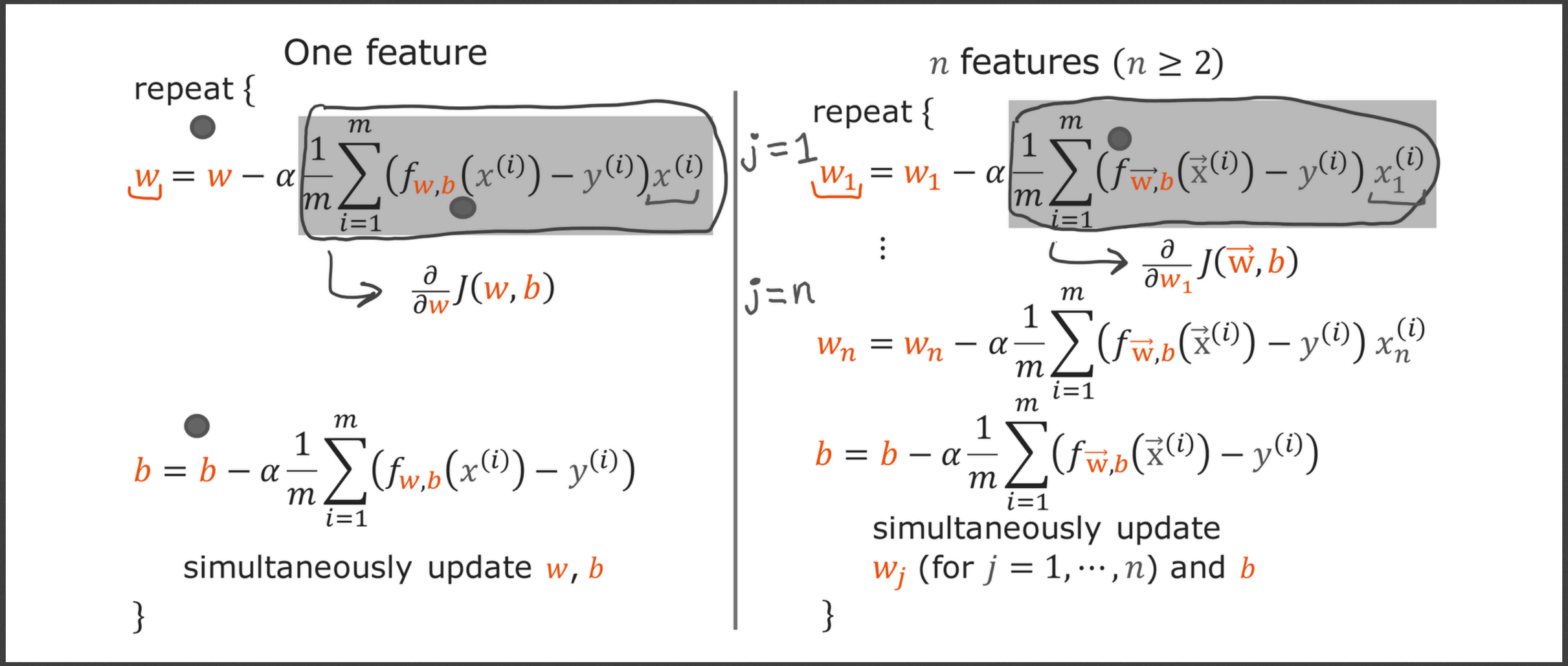
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Gradient Descent with Many Features

The Same Model in Vector Notation

	Previous notation	Vector notation
Parameters	$w_1, \cdots, w_n$ b	$\vec{w} = [w_1 \quad \cdots \quad w_n]$ ← vector of length n b still a number
Model	$f_{\vec{w},b}(\vec{x}) = w_1 x_1 + \cdots + w_n x_n + b$	$f_{\vec{w},b}(\vec{x}) = \vec{w} \cdot \vec{x} + b$ ← dot product
Cost function	$J(w_1, \cdots, w_n, b)$	$J(\vec{w}, b)$
Gradient descent	repeat {	repeat {
	$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(w_1, \cdots, w_n, b)$	$w_j = w_j - \alpha \frac{\partial}{\partial w_j} J(\vec{w}, b)$
	$b = b - \alpha \frac{\partial}{\partial b} J(w_1, \cdots, w_n, b)$	$b = b - \alpha \frac{\partial}{\partial b} J(\vec{w}, b)$
	}	}

Updating Every Feature Weight



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Why Feature Scaling Helps

Feature Size and Weight Size

$\widehat{price} = w_1 x_1 + w_2 x_2 + b$

size # bedrooms

x_1: size (feet^2)
range: $300 - 2{,}000$

large

x_2: # bedrooms
range: $0 - 5$

small

House: $x_1 = 2000$, $x_2 = 5$, $price = \$500k$

one training example

size of the parameters w_1, w_2?

$w_1 = 50$, $w_2 = 0.1$, $b = 50$

$\widehat{price} = \underbrace{50 * 2000}_{100{,}000k} + \underbrace{0.1 * 5}_{0.5k} + \underbrace{50}_{50k}$

$\widehat{price} = \$100{,}050.5k = \$100{,}050{,}500$

$w_1 = 0.1$, $w_2 = 50$, $b = 50$

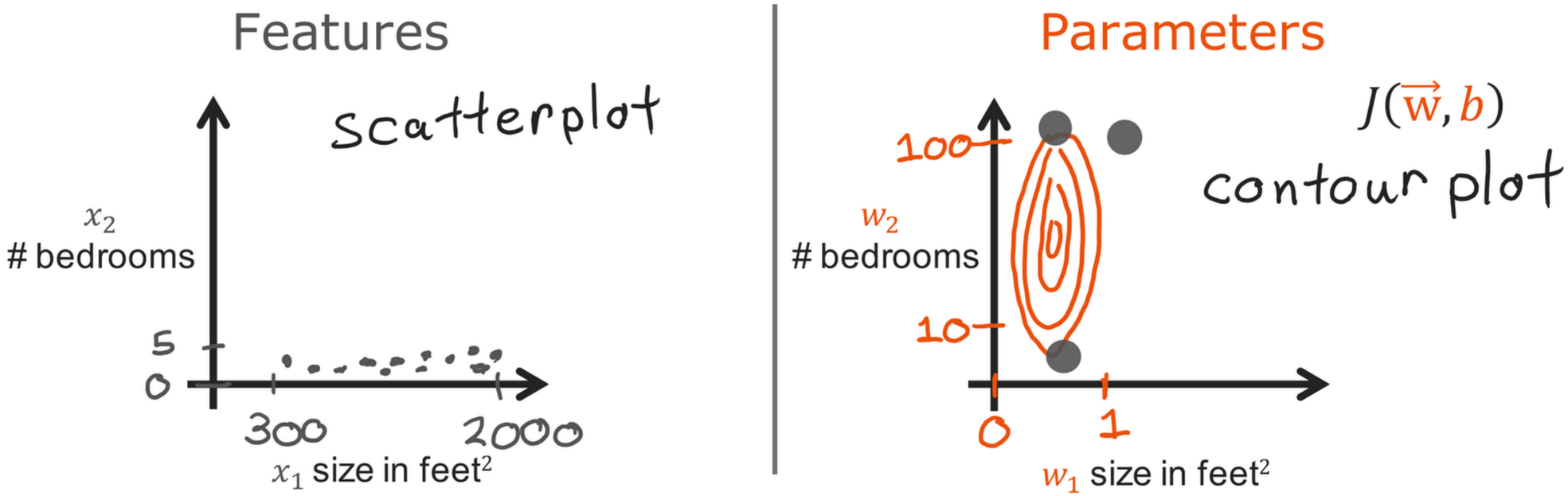
small large

$\widehat{price} = \underbrace{0.1 * 2000k}_{200k} + \underbrace{50 * 5}_{250k} + \underbrace{50}_{50k}$

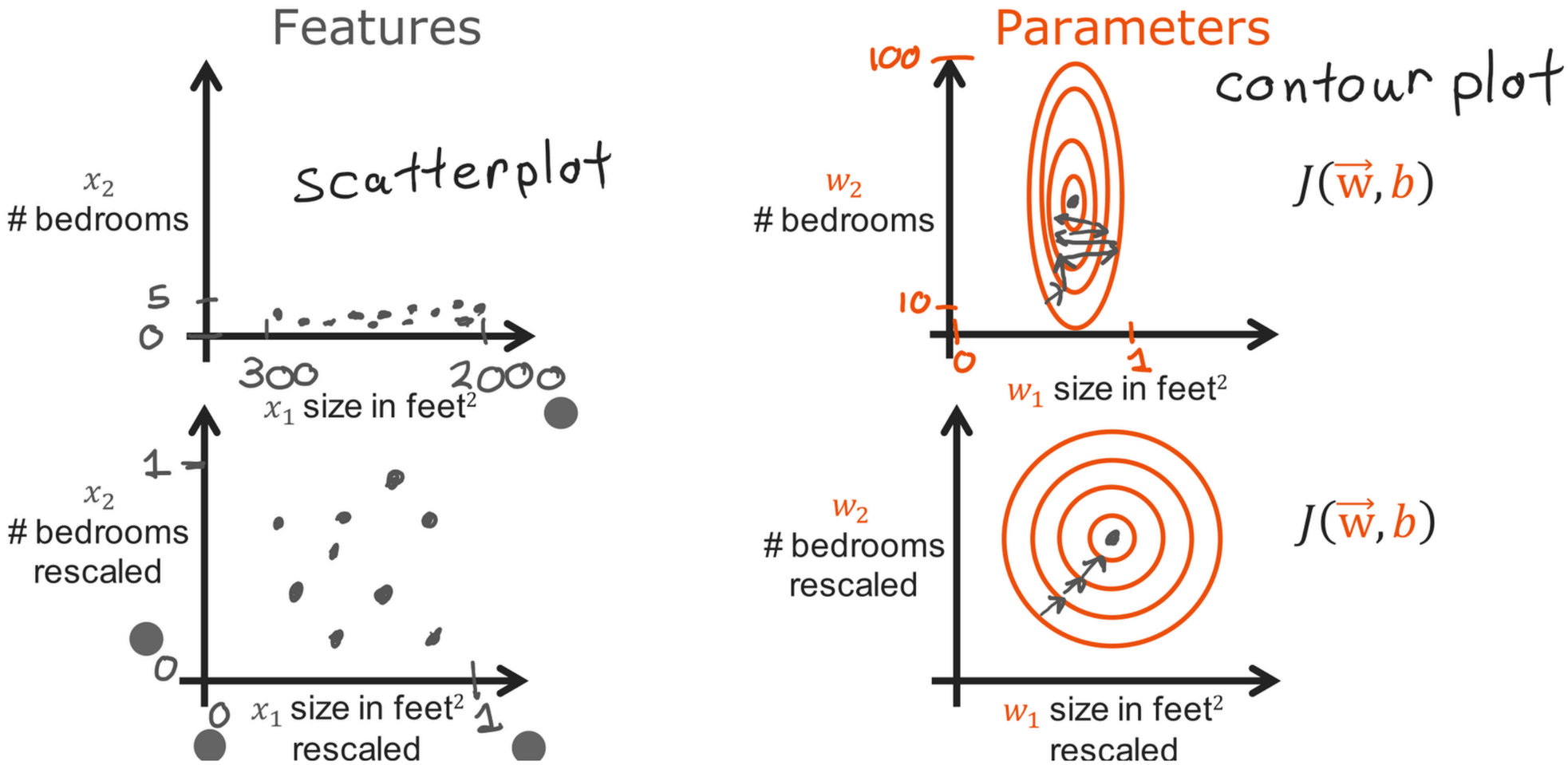
$\widehat{price} = \$500k$ more reasonable

Why the Cost Surface Gets Stretched

	size of feature x_j	size of parameter w_j
size in feet^2	⟷	⟷
#bedrooms	⟷	⟷



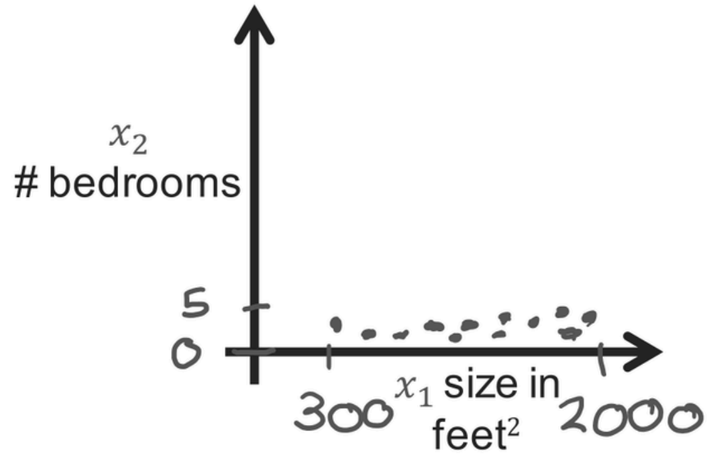
Scaling Helps Gradient Descent

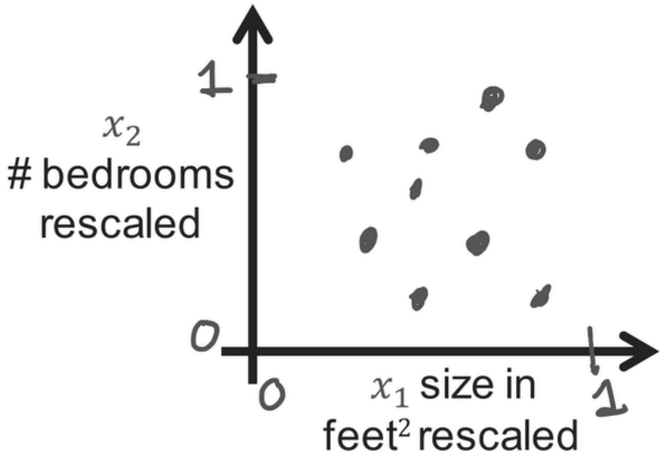


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How to Scale Features

Dividing by Each Feature's Maximum



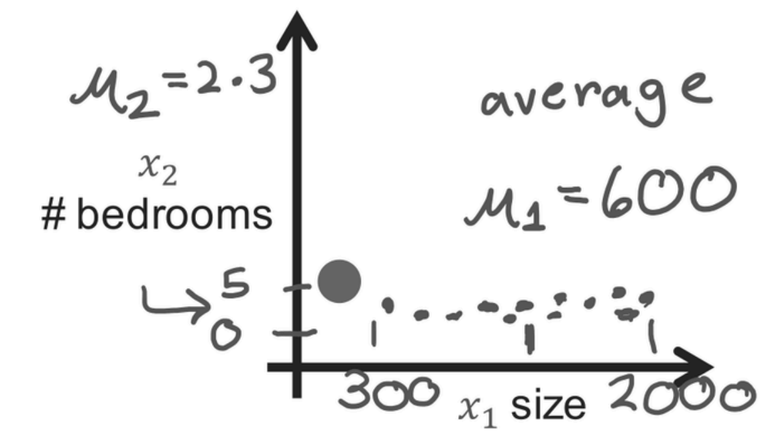


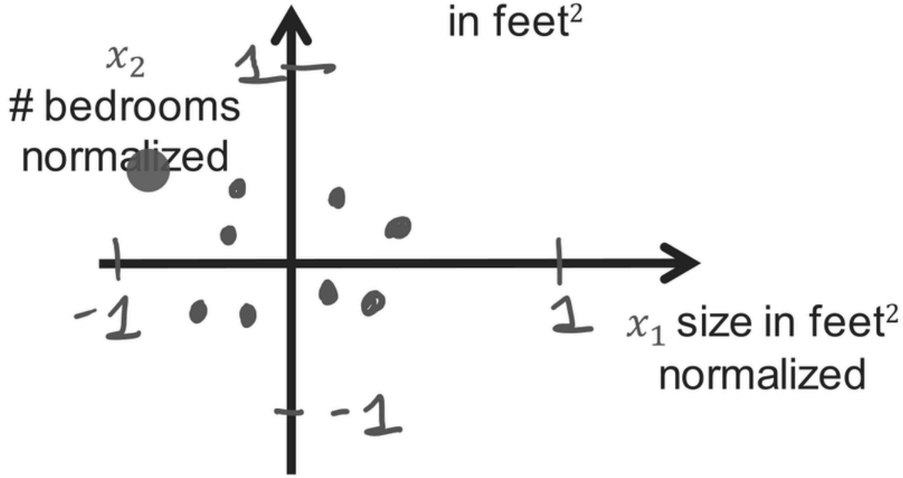
$$300 \leq x_1 \leq 2000 \qquad 0 \leq x_2 \leq 5$$

$$x_{1,scaled} = \frac{x_1}{2000} \quad \text{max} \qquad x_{2,scaled} = \frac{x_2}{5} \quad \text{max}$$

$$0.15 \leq x_{1,scaled} \leq 1 \qquad 0 \leq x_{2,scaled} \leq 1$$

Centering Features around the Mean





$$300 \le x_1 \le 2000$$

$$x_1 = \frac{x_1 - \mu_1}{2000 - 300}$$

max-min

$$-0.18 \le x_1 \le 0.82$$

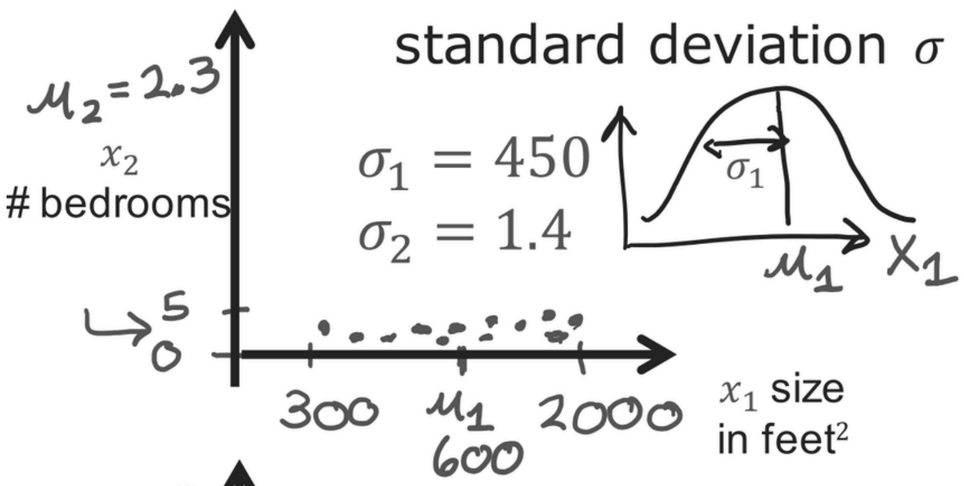
$$0 \le x_2 \le 5$$

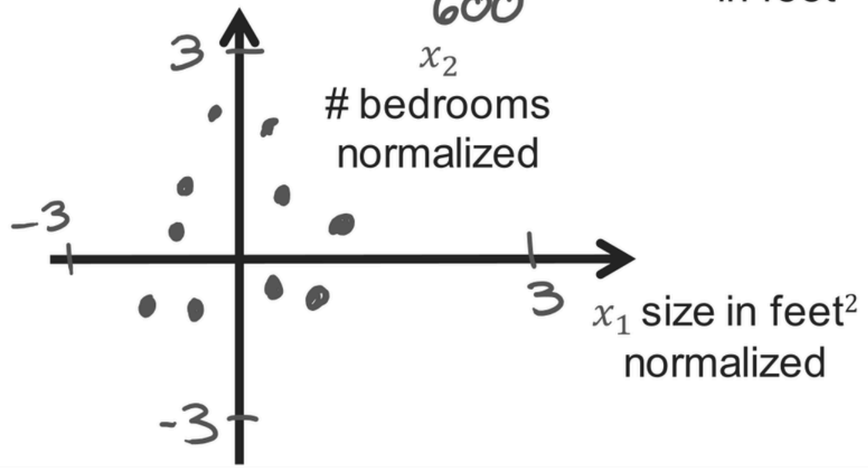
$$x_2 = \frac{x_2 - \mu_2}{5 - 0}$$

max-min

$$-0.46 \le x_2 \le 0.54$$

Scaling by the Standard Deviation





$$300 \le x_1 \le 2000 \qquad 0 \le x_2 \le 5$$

$$x_1 = \frac{x_1 - \mu_1}{\sigma_1} \qquad x_2 = \frac{x_2 - \mu_2}{\sigma_2}$$

$$-0.67 \le x_1 \le 3.1 \qquad -1.6 \le x_2 \le 1.9$$

Choosing Similar Feature Ranges

aim for about $-1 \leq x_j \leq 1$ for each feature x_j

$-3 \leq x_j \leq 3$

$-0.3 \leq x_j \leq 0.3$

} acceptable ranges

$0 \leq x_1 \leq 3$ okay, no rescaling

$-2 \leq x_2 \leq 0.5$ okay, no rescaling

$-100 \leq x_3 \leq 100$ too large → rescale

$-0.001 \leq x_4 \leq 0.001$ too small → rescale

$98.6 \leq x_5 \leq 105$ too large → rescale